

Overcoming Dataset Imbalances and Computational Challenges in IoT Intrusion Detection: A SMOTE-Enhanced Transformer-Based Model

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Abstract— Due to security holes in IoT-enabled networks, the massive adoption of IoT has unleashed a new wave of threats into computer networks. Due to their limited resources, these devices are ill-equipped to handle security measures, making them the most vulnerable part of our computer networks and putting our data and systems at risk. Some have suggested Intrusion Detection Systems (IDS) as a means to lessen the impact of intrusions connected to the IoT in an effort to solve this problem. Although IDS are essential for detecting threats, they are not widely used because of their high computing costs and reliance on labelled data. Finding a happy medium between fast detection and excellent accuracy is the objective of this model development effort. In order to accomplish this, to present a feature selection method based on the black-winged kite algorithm (BKA), a groundbreaking swarm intelligence technique that combines the Leader tactic and the Cauchy variation procedure to find the expansive appropriate convergence solution. BKA is based on the migratory and predatory instincts of black-winged kites. The study used few-shot learning techniques by leveraging transfer learning models and specifically using pre-trained architectures for categorisation. It also fine-tuned four pre-trained transformers. To take into account the imbalance in these datasets as a significant component and use the Synthetic Minority Over-sampling Technique (SMOTE) to address it. The research team used BoT-IoT, TON-IoT, and CIC-DDoS2019 as benchmark datasets to assess the impact of the chosen features on the suggested model's performance. Metrics for evaluation include F1-score, recall, precision, and accuracy of detection. Evidence from experiments shows that the suggested ensemble model outperforms state-of-the-art AIDS topologies for IoT networks in terms of detection accuracy and efficiency.

Keywords— *Synthetic Technique; Intrusion Detection Systems; Internet of Things; Black-winged kite algorithm; Transformer Model*

I. INTRODUCTION

When hackers are able to break into a system and steal, edit, or delete data, it can harm other systems connected to the network [1]. This is known as a system vulnerability. To urgently want effective security and data privacy procedures to address future difficulties, as various attacks endanger the normal running of networks and schemes [2]. In order to address security concerns while building the Internet,

efficient dynamic traffic management of networks is essential. Monitoring network presentation, precisely comprehending network dynamics and operations, addressing most user needs, and assuring quality of service (QoS) are some of the major characteristics that must be considered while building any security solution [3]. The CIA triad—confidentiality, integrity, and availability—must be prioritized by security engineers while providing network system security in order to guarantee improved and consistent performance [4].

When it comes to security threats and challenges, the research community's ultimate concentration is on intrusion detection systems (IDSs). These systems have lately offered helpful answers to the problems posed by harmful actions [5]. Intrusion Detection Systems (IDSs) can be host-based (software-based) or network-based (hardware-based) and can identify malicious activity on either the host or the network. Different intrusion detection systems are based on different criteria, such as signatures, specifications, or anomalies [6]. In the first, a set of criteria is used to monitor incoming network packets for skewness; in the second, traffic flow is compared to stored signatures of known assaults; and in the third, network threats are identified [7]. One major problem of specification-based IDS is the need for specialist knowledge, while signature-based IDS requires constant database changes to keep up with the ever-increasing number of attacks [8]. The third kind is more sophisticated than the previous two and mostly employs classification techniques to resolve issues. Much effort has been done on intrusion detection, according to the review of existing systems [9]. If to want to improve the accuracy of attack classification, to should use AI and ML techniques [10]. In terms of efficiently and effectively distinguishing between normal and attack cases, a number of approaches based performed better. Finding the many forms of assaults that could compromise the network is a common task for most.

Cyberspace is becoming increasingly important due to the proliferation of internationally connected gadgets brought about by the fast evolution of internet and information technology [11]. Cyberattacks, which risk private information, company records, and even national security,

have grown in sophistication and frequently alongside technological advancements. The four main types of traditional intrusion detection systems (IDSs)—rule-based, signature-based, misuse-based, and anomaly-based—have all seen extensive deployment in the fight against network security breaches [12]. Although these systems are effective against known assaults, they have serious limitations when it comes to detecting new or changing threats because they rely on predetermined rule sets or features that are manually retrieved [13].

Researchers in the area of cybersecurity have recently attempted to improve intrusion detection using automated feature learning by applying deep learning technologies, which have been rapidly advancing, to the subject. While Recurrent Neural Networks (RNNs) are great at capturing temporal dependencies, Convolutional Neural Networks (CNNs) are great at identifying spatial patterns in network flows [14]. In situations when there is an abundance of labelled data, deep learning algorithms excel, but when there is a lack of labelled data, they frequently fail to generalize. For instance, conventional deep learning models have a hard time accurately capturing the features of malicious traffic because labelled harmful traffic data is scarce and varied, in contrast to the majority of normal data in real-world network traffic [15]. On top of that, models trained on historical data rapidly become useless when faced with new types of attacks. For example, APTs that emerge after 2020 and supply chain attacks that use old models have shown inadequate resilience [16].

One potential approach to these problems is few-shot learning. Matching networks and prototypical networks are two techniques that have demonstrated promise in using small samples to identify unknown assaults. In high-dimensional and dynamic network traffic circumstances, however, traditional few-shot learning approaches show limits [17]. Nevertheless, there is still no clear solution to the research question of how to enhance resilience against unexpected threats by combining inductive learning with improved feature extraction techniques.

- o Developing a reliable Intrusion Detection System for IoT networks is the main goal of this research. To do this, to will use the Black-Winged Kite Algorithm (BKA) to improve model performance and convergence, and to will also design a feature selection strategy.

- o To tackle the problems of limited labelled data and high computing costs, to are using few-shot learning techniques and fine-tuned pre-trained transformers. To make our model more reliable and resilient, to are using the SMOTE to fix dataset imbalances.

- o Applying presentation criteria including detection accuracy, precision, recall, and F1-score to benchmark IoT datasets (BoT-IoT, TON-IoT, and CIC-DDoS2019) in order to assess the suggested model.

- o Outperforming current IDS designs for IoT networks, the suggested model proves its superiority by

striking a balance between fast detection and accurate detection.

II. RELATED WORKS

A new Intrusion Detection System (NIDS) for IoT built on Graph Neural Networks (GNNs) was presented by Lin et al. [18]. Graph-based data's topological structure is used by GNNs to construct traffic flow correlations. In a flow-based representation, the data can build the network structure of a graph using the addresses of its sources and destinations. In order to enhance the sampling embeddings, this research presents a new and enhanced GNN, GraphSAGE, which is an edge-based graph neural network model. The GNN incorporates global attention and local gating mechanisms. The next step in accomplishing IoT network intrusion detection is to apply graph contrastive learning to improve feature representation. This will allow for a more thorough capture of graph edge features and topological information. To refer to this approach as E-GRACL, which stands for Edge-based contrastive learning. To address the issue of intrusion detection in IoT networks using flow-based data by applying GNN. The results show that our method has certain benefits in both metrics, indicating the enormous potential of GNN in IoT intrusion detection.

An intrusion detection system with three stages, suggested by Karuppiah et al. [19], can detect and prevent assaults on pervasive networks by means of a predictive machine learning model. Stage one involved classifying network flows using AARM feature selection with support vector machines (SVMs). Second, a behaviour analysis model based on ensemble learning (PAEL) was proposed. Our last proposal was a PAEL model that relies on swarm bat optimization for the classification of labels used in multiple tasks. To achieve lower resource utilization and increase system efficiency, the trained model is deployed to fog and edge computing devices. First, anomalies in network traffic from IoT devices are detected at the edge devices; second, a sample of this data is sent to the fog computing device to confirm the attacks and abnormalities; and finally, the cloud server receives the final sample of identified data. Training on sample data, including newly found samples, allows the proposed PML-GWEL model to continuously increase its accuracy. The approach is based on predictive machine learning (ML). All nodes in the network get the trained model after it's finished training, so they may update their main detector models and get rid of any old pre-detector models. This procedure enhances the system's overall efficiency by decreasing the hardware resources utilized by the pre-detector representations. To compare the suggested model to various methods that are already in use.

The XI2S-IDS architecture, developed by Mahmoud et al. [20], is an intelligent two-stage intrusion detection system that can be explained. For low-frequency attacks, the XI2S-IDS architecture provides better detection and interpretability by combining a two-stage technique with SHAP-based

explanations. First, to do binary classification; second, to go on to multi-class classification. Through the utilization of SHAP values, XI2S-IDS improves decision-making transparency, granting security analysts clear insights into the significance of features and the reasoning behind the model. Results show that experimental results on the UNSW-NB15 and CICIDS2017 datasets significantly enhance detection performance, with a considerable decrease in false negative attacks and good precision, recall, and F1-scores maintained.

Using TPE-FL-LightGBM and an enhanced binary simulated annealing technique (IBSA), Luo et al. [21] present a new approach to network intrusion detection. First, to implement cost-sensitive learning by including Focal Loss into function. This makes the model better at learning from data that are hard to classify and reduces the effect of class imbalance on its performance. The simulated annealing process is then modified to include adaptive initial temperature and Metropolis criterion changes, multi-neighborhood search strategies, and an S-shaped transfer function, among other enhancements. Thanks to these updates, the IBSA approach can now choose optimal features efficiently with less iterations. Lastly, the FL-LightGBM classifier is made even more effective by optimizing its structure using the Tree-structured technique. The suggested network intrusion detection approach is proven reliable through extensive experiments on the UNSW-NB15 and NSL-KDD datasets.

To accurately categorize network intrusions, Roy, & Kalita [22] presented an intelligent semi-supervised intrusion detection system that relies on various algorithms. The first step of the pre-processing is to normalize the standard scaler and delete any null values. Following the pre-processing phase, an EDRL model is used to learn intricate patterns from data through environmental interaction and extract high-level representations. By combining a deep autoencoder (AE) with an improved flamingo search algorithm (IFSA), deep reinforcement learning is enhanced to approximating the Q-function and selecting the optimal policy. A support vector machine (SVM) classifier is for classification after feature representations. It can distinguish between normal and attack instances in the input. Using the UNSW-NB15, CICIDS2017, and NSL-KDD datasets, the given model is tested in a Python platform simulation. Outperforming the current detection frameworks, the general classification accuracy is 99.6% when using the UNSW-NB15 dataset, 99.93% when using the CICIDS2017 dataset, and 99.42% when using the NSL-KDD dataset.

III. PROPOSED METHODOLOGY

In this section, the detection of network traffic is carried out by proposed deep learning model and it is visually shown in Figure 1, where each of its blocks are discussed in the upcoming sections.

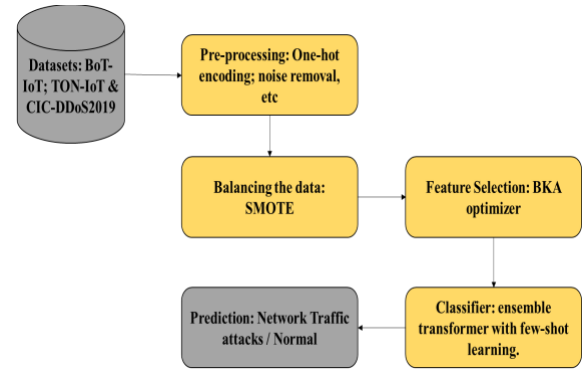


Figure 1: Workflow of the Research Model

A. Dataset specification

Using tagged datasets of network traffic helps the supervised models perform better. To achieve this goal, to must supply AIDS with the necessary data so that it can be trained to reliably and accurately detect an enormous number of network threats. Every day, more and more Internet of Things (IoT) network, which results in an explosion of unstructured data. Our suggested approach was developed using a dataset generated from a genuine setting in order to ensure efficient operation in real-time [23]. The range of assaults included in this dataset guarantees that our model can correctly identify and categorise various forms of incursions. This dataset will be useful for training our model to detect features and patterns in real-time network traffic. In recent times, a plethora of new flow-based benchmark datasets have been made available, including BoT-IoT [24], TON-IoT [25], and CIC-DDoS2019 [26]. For effective anomaly-based network intrusion detection, to employ these datasets as up-to-date and realistic data from network traffic. You may find a summary of these datasets' key features in Table 1.

Table 1: Explanation of BoT-IoT, TON-IoT, besides CIC-DDoS2019 datasets.

Datasets	Size		No. of features	Attack label
	Normal	Attack		
Bot-IoT [24]	477	3,668,045	46 features	DoSTCP, DoSUDP, DDoSHTTP, DDoSUDP, OS_Fingerprint, Facility scan, Statistics Exfiltration, Keylogging
TON-IoT [25]	500,000	2,500,000	45 features	DoS, DDoS, MITM, Scanning, XSS
CIC-DDoS2019 [26]	94,568	331,354	79 features	NTP, DNS, MSSQL, NetBIOS, SNMP, UDP, UDP-Lag, WebDDoS, TFTP, Portmap

1) *BoT-IoT dataset*

An authentic network environment was constructed in the Cyber Range Lab at UNSW Canberra to establish the BoT-IoT dataset. Refrigerator, garage door, thermostat, light, and weather monitoring system data was among the many items contained in this. As shown in Table 1, the trials utilised 5%, or 3.6 million records, of the overall dataset of 72 million records.

2) *TON-IoT dataset*

We used three publicly available benchmark datasets for IoT intrusion detection: BoT-IoT, TON-IoT, and CIC-DDoS2019. These datasets were selected because they cover a wide range of network threats, ensuring robustness in evaluating our model.

BoT-IoT Dataset:

This dataset was generated in a real IoT environment at UNSW Canberra and contains both normal and attack traffic. The dataset consists of 3,668,045 attack samples and 477,000 normal samples across 46 extracted features. Collected from IoT network telemetry, this dataset has 500,000 normal samples and 2.5 million attack samples, with 45 extracted features. Attack types include MITM, scanning, DoS, DDoS, and XSS.

Attack categories include DoS TCP, DoS UDP, DDoS, OS fingerprinting, keylogging, and reconnaissance scans. The Cyber IoT Lab at UNSW Canberra built a realistic and enormous-scope network, which was used to produce the TON-IoT dataset. Information gathered from the Internet of Things (IoT) includes telemetry data, traffic. Our experiments made advantage of data collected from the network's traffic. The tests utilised 7%, or 3 million records, of the 22,339,021 entries contained in the TON-IoT dataset. As seen in Table 1, several assaults that did not impact the IoT network were not included in the study.

3) *CIC-DDoS2019 dataset*

The Canadian Institute for Cybersecurity's CIC-DDoS dataset comprises a wide variety of distributed denial of service (DDoS) assaults. It was gathered on two different days for the sake of training and testing. Twelve distinct DDoS assault types are included in the training set, which was taken on January 12th, 2019.

- Developed by the Canadian Institute for Cybersecurity, this dataset focuses on DDoS attacks on multiple protocols.
- It consists of 94,568 normal samples and 331,354 attack samples, with 79 extracted features.
- The attack types include NTP, DNS, MSSQL, NetBIOS, SNMP, UDP, UDP-Lag, WebDDoS, TFTP, and Portmap.

The following protocols are susceptible to attacks: NTP, LDAP, DNS, MSSQL, TFTP. In contrast, seven different attacks—NetBIOS, MSSQL, PortScan, LDAP, UDP-Lag, UDP, besides SYN—make up the test dataset.

B. *Data preprocessing*

The term "data preprocessing" refers to any process done on it ready to be used in another data processing operation. Part of getting ready to work with data is this. It is a crucial initial step in data mining and has always been. In order to build, train, and execute inferences against ML and AI models, data preparation methodologies have been substantially improved.

❖ *Data Cleaning:*

The duplicate records from the datasets are removed to prevent redundancy. Any corrupted or incomplete data entries (e.g., missing values, null values) are handled using imputation techniques. Missing numerical values are filled using the mean of the respective feature, while categorical missing values are replaced using the most frequent value.

❖ *Data Normalization:*

Since the datasets contain features with varying scales, we apply Z-score normalization (also known as standardization) to transform numerical values into a standard normal distribution. The transformation follows the equation:

Data preprocessing is use in data science initiatives like data mining and machine learning by making it more easily analyzed and managed. In order to ensure accurate findings, the methods are typically used at the start of the ML and AI pipeline. The computer-generated structured data files (CSVs) containing the dataset are accessible to the public. Using sk-learn's Standard Scaler() approach, the data was normalized. It scales the features down to unit variance and subtracts the mean to standardize them. Data pretreatment encompasses all activities performed on data to make it ready for usage in data processing processes that follow. Before importing datasets into Google COLAB in CSV format, it has always been an integral aspect of data preparation. Deleted, duplicated, or otherwise unusable data, as well as values for data balancing.

The preprocessed dataset is divided into training and testing subsets. We use two configurations:

- 70%-30% split: 70% for training, 30% for testing
- 60%-40% split: 60% for training, 40% for testing

Also, in order to provide ML algorithms with numerical representations of categorical values, the one-hot encoding method is used. This method creates a feature column for every possible category in a categorical feature. Since ML does not support categorical values, the one-hot encoding technique's categorical data transformation converts each category in a categorical feature into its own binary vector.

C. *Data balancing*

To start a new phase of data preprocessing, to will use the SMOTE technique to balance the dataset. One way to create artificial minority class samples is outlined in the SMOTE algorithm. When data problems, SMOTE is a widely used oversampling approach. It replicates minority-class situations at random in an effort to level the distribution of classes. New minority instances are shaped by SMOTE by existing minority instances. An enhanced approach to handling data imbalances in classification contexts is the SMOTE algorithm. The observed frequencies among the various possible outcomes of a categorical variable are markedly distinct, as shown by an unbalanced dataset. The main thing is that most of the observations are of one type and very few of another. Aiming to achieve dataset parity:

- The process of random oversampling involves taking data from a minority class and either adding it to the training sample or replacing it with it.
- Under-sampling or re-sampling indicates the opposite of random oversampling. The process entails selecting data at random from the majority of the class and either adding it to the training dataset or replacing it.
- The SMOTE algorithm is a method that helps with ML classification and is used with datasets that are not balanced. It entails augmenting the dataset with afresh created data derived from existing minority data.

Because the quality of the data has a impact on how well ML models work, data preparation is a crucial part of the ML process. Data preparation involves updating and purifying the dataset so it fits the ML model.

D. Feature selection using Black-winged kite algorithm (BKA)

In this study, to feed BKA [27] balanced data so it can pick out the most important attributes to boost classification accuracy. The BKA combines the Leader Cauchy variation procedure to increase the intensity of aggregate discovery and find the right computational solution. This not only mimics the fabulous adaptability of black-winged kites to change their environment and target locations, but it also captures their migratory and predatory instincts.

1) Initialization populace

The populace is haphazardly primed, besides a matrix BK is specified as:

$$BK = \begin{bmatrix} BK_{1,1} & BK_{1,2} & \cdots & BK_{1,dim} \\ BK_{2,1} & BK_{2,2} & \cdots & BK_{2,dim} \\ \vdots & \vdots & \ddots & \vdots \\ BK_{pop,1} & BK_{pop,2} & \cdots & BK_{pop,dim} \end{bmatrix} \quad (1)$$

Where pop showcases the populace magnitude, pop dimension, $BK_{i,j}$ cabinetsjth dimension of ith individual. The X_i is specified as:

$$X_i = BK_{lb} + rand(BK_{ub} - BK_{lb}) \quad (2)$$

Wherei showcases annumeral in $[1, pop]$, BK_{lb} besides BK_{ub} showcase the limitations, and $rand \in [0, 1]$.

The most fitting leader X_L is specified as:

$$f_{best} = \min (f(X_i)) \quad (3)$$

$$X_L = X(find(f_{best} == f(X_i))) \quad (4)$$

2) Assaulting behavior

Black-winged kites hunt parasites and small animals in pastures by swooping down on their prey and controlling the angle of their wings and tail in relation to their speed. It is specified that the aggressive conduct must be:

$$y_{t+1}^{i,j} = \begin{cases} y_t^{i,j} + n(1 - \sin(r)) \times y_t^{i,j} & p < r \\ y_t^{i,j} + n(2r - 1) \times y_t^{i,j} & else \end{cases} \quad (5)$$

$$n = 0.05 \times e^{2 \times (\frac{t}{T})^2} \quad (6)$$

Where $y_t^{i,j}$ and $y_{t+1}^{i,j}$ showcase the sites of dimension, $r \in [0, 1]$, $p = 0.9$, T repetition termination.

3) Migration behavior

Food availability and humidity are two of several environmental factors that limit the complex behaviour of migrating birds. As a means of adjusting to the changing seasons, many black-winged kites travel south during winter in search of better living conditions and more material supplies. The dominant species will decide to revert to its migratory state if its expected fitness is lower than that of the random species. On the flip side, the group's leader will provide instructions to the searchers until they reach their target. To ensure a successful migration, this method consistently identifies outstanding leaders.

The behaviour of migration is defined as

$$y_{t+1}^{i,j} = \begin{cases} y_t^{i,j} + C(0,1) \times (y_t^{i,j} - L_t^j) F_i < F_{ri} \\ y_t^{i,j} + C(0,1) \times (L_t^i - m \times y_t^{i,j}) & else \end{cases} \quad (7)$$

$$m = 2 \times \sin(r + \pi/2) \quad (8)$$

where L_t^j scorer, F_i current site, F_{ri} accidental site, $C(0, 1)$ showcases the Cauchy mutation.

The Cauchy is defined as a continuous two-metric stochastic distribution that is:

$$f(x, \delta, \mu) = \frac{1}{\pi} \frac{\delta}{\delta^2 + (x - \mu)^2} - \infty < x < \infty \quad (9)$$

where $\delta = 1$, $\mu = 0$, the likelihood density fitness is specified as:

$$f(x, \delta, \mu) = \frac{1}{\pi} \frac{\delta}{x^2 + 1} - \infty < x < \infty \quad (10)$$

Algorithm 1 Black-winged kite algorithm

Begin

Step 1. Initialize the black-winged kite population $X_i (i = 1, 2, \dots, n)$.

Step 2. Ascertain the fitness value of kite.

- Obtain X_{best} and F_{best} .

Step 3. while (t<T) do

/* Attacking phase */

if $p < r$

Refresh the attacking position via Eq. (5):

$$y_{t+1}^{i,j} = y_t^{i,j} + (1 + \sin(r)) \times y_t^{i,j}$$

else if do

Refresh attacking position via Eq. (5):

$$y_{t+1}^{i,j} = y_t^{i,j} + n(2r - 1) \times y_t^{i,j}$$

end if

/* Migration phase */

if $F_i < F_{ri}$

Refresh migration position via Eq. (7):

$$y_{t+1}^{i,j} = y_t^{i,j} + C(0,1) \times (y_t^{i,j} - L_t^j)$$

else if do

Refresh migration position via Eq. (7):

$$y_{t+1}^{i,j} = y_t^{i,j} + C(0,1) \times (L_t^j - m \times y_t^{i,j})$$

end if

/* Select the best individual */is

if $y_{t+1}^{i,j} < L_t^i$

$$X_{best} = y_{t+1}^{i,j}, F_{best} = f(y_{t+1}^{i,j})$$

else if do

(end condition logic appears here)

end if

/* Select the best individual */

if $y_{t+1}^{j,i} < L_t^i$

$$X_{best} = y_{t+1}^{j,i}, F_{best} = f(y_{t+1}^{j,i})$$

else if do

$$X_{best} = L_t^i, F_{best} = f(L_t^i)$$

end if

Boundary control and modification.

Ascertain the fitness value of kite.

Alter X_{best} if there is a superior black-winged kite.

$$t = t + 1$$

end while

Return X_{best} and F_{best}

End

E. Classification Models

Here to will go over what transformers are and why their fine-tuning function is beneficial. Also included is a synopsis of the models that formed the basis of this study.

1) Fine-Tuning

Transformer models are state-of-the-art AI systems that can comprehend, harvest, and alter human speech. They

gather and analyze text data using deep learning architectures, particularly transformer models. Text categorization, sentiment analysis, translation, and text summarization are just a few of the many popular jobs that transformers can accomplish swiftly. Due to their superior modelling of the contextual relations provided words or phrases, transformers can be incredibly effective in text categorization tasks compared to more conventional methods. This is mostly due to the fact that these models can swiftly and easily adjust to new jobs with minimal adjustments, as knowledge is gained from a large amount of diverse text data. To fine-tune them, one must first adapt them to a certain domain, then improve their performance on tasks and terms related to that domain. In addition to reducing biases, this procedure disclosures the model to more diverse and representative data, which increases accuracy and relevance in predictions and responses. Four representations—BERT, DistilBERT, RoBERTa, and XLNET—were fine-tuned in this study.

An innovative paradigm that popularized the use of transformers is BERT, which Transformers. To be more specific, BERT considers both contexts simultaneously, as opposed to constitutive models that only process the text in one direction. A deeper understanding of the languages and their word relationships is the result. Using masked language modelling prediction, BERT is trained on a huge text corpus. After being applied to several text categorization tasks with success, the model evolved into a state- model.

Knowledge distillation allowed us to design DistilBERT, a more compact, quicker, and lighter variant of BERT. It is 60% quicker and uses less memory to operate, while preserving 97% of BERT's language understanding capabilities. When dealing resources, DistilBERT becomes an attractive choice. Although it is smaller, DistilBERT excels in text classification and many other natural language processing tasks. To enhance the performance of our intrusion detection system, we fine-tuned four transformer models—BERT, DistilBERT, RoBERTa, and XLNet—on the BoT-IoT, TON-IoT, and CIC-DDoS2019 datasets. Fine-tuning involves updating the pre-trained weights of the model to adapt to a domain-specific task, in this case, intrusion detection in IoT networks

In Learning Rate 2e-5 for BERT, RoBERTa, XLNet; 3e-5 for DistilBERT lower learning rate (2e-5) ensures stable convergence while allowing small updates to pre-trained weights. DistilBERT, being a smaller model, requires a slightly higher rate (3e-5) for better adaptation.

Batch Size 16 and 32 experimented with batch sizes of 16 and 32 to balance computational efficiency and gradient stability. A batch size of 16 was ideal for avoiding memory issues on larger datasets, while 32 was preferred for smaller datasets.

An improvement over BERT is RoBERTa, which stands for A Robustly Optimized BERT Pretraining Approach. Designed by Facebook AI researchers, it modifies the BERT model's architecture and training process to

increase the model's capabilities. To achieve this, to trained on a bigger dataset with longer sequences and removed the next sentence prediction target. In order to understand contextual representation, it employs dynamic masking during training as well. Thanks to these enhancements, RoBERTa now achieves state-of-the-art results in text classification tasks besides dominates several benchmarks, such as GLUE, RACE, and SQuAD. After BERT, RoBERTa outperforms it in accurately classifying and comprehending complicated textual material by better capturing textual context and linkages.

By taking use of bidirectional context and rearranging input arrangements, XLNet, a pre-trained classical, successfully combines the advantages of BERT with autoregressive representations such as GPT. This gives XLNet a more nuanced and versatile grasp of word relationships by allowing it to record all conceivable word orders. Specifically, the model's adaptability allows it to better grasp text classification tasks' meaning and contextual relationships, leading to significantly improved accuracy and performance.

2) Few-Shot Learning

To also used few-shot learning in this work. The model can accurately understand a task with just a few instances of a certain class, according to this machine-learning paradigm. This comes in handy especially when dealing with impractically large datasets. This technique has been recently integrated into LLMs by the training of a model on the majority of the text, enabling it to accomplish several jobs with minimal input. The goal of few-shot learning is to train models to generalize using a small sum of training examples. This method is particularly useful in vision, but it can be used to many other areas as well. Start with a pre-trained classical, typically a transformer language model (BERT, GPT, DistilBERT), for few-shot fine-tuning. This is in stark contrast to traditional fine-tuning methods, which typically rely on access to massive labelled datasets in order to achieve satisfactory results. To accomplish this by making use of the pre-trained model's knowledge, which enables it to generate well-informed predictions given only a small number of samples.

In this study, to utilized SetFit, a framework that integrates fine-tuning with pre-trained big language models to enhance few-shot learning for text categorization problems. There are two main components to the framework's operation: first, it uses a pre-trained language model to encode text; and second, it generates embeddings for both the text task-specific collection of labels. In order to train the model to capture clarity required for classification, the embeddings are trained on a limited sum of labelled examples. It makes use of contrastive learning, a method wherein the classical is taught to differentiate among instances in the embedding space that are comparable and those that are different by identifying small differences across classes according to their relationships. On the other hand, certain methods may rely on a plethora of labels without

ensuring optimal performance. Considering that many real-world situations have limited labelled data, this is highly efficient in practice. SetFit achieves great consequences for classification tasks like sentiment topic categorization using pre-trained models mixed with limited forecasting. This alleviates the normal data necessities besides computing expenses LLMs.

3) Explainability

Worries regarding the inner workings of AI systems have grown in tandem with their level of complexity. An increasingly popular solution to this problem is the concept of explainable AI. Explainability in artificial intelligence (AI) is the process of making AI systems' actions and results comprehensible to people. The goal is to shed light on how AI models draw conclusions; this development promotes openness, responsibility, and trust. Conversely, Large Language Models can effectively understand and generate human languages since they are trained on massively scaled datasets that comprise complicated neural architectures. But the amount of data they handle and the intricacy of their design make explainability difficult. A thorough understanding of transformer operation is critical for a number of reasons. Users' faith in the models and the results they produce is the primary concern of the first bullet item. To avoid biases, the model can help find imbalances in the training data and fix them, according to the second point. Attention visualization, a tool for studying the operation of attention apparatuses in neural networks, particularly in representations like LLMs, was employed in this study. When making predictions or producing outputs, attention processes enable models to prioritize various input aspects. Models can generate responses or predictions by focusing on specific words sequence through attention. These mechanisms, which can be illustrated graphically, are helpful for comprehending the system's processes since they specify the input regions that are suitable for a specific job. The majority of attention concentration demonstrations make use of maps, which show the intensity of focus on specific input tokens. As an example, the model may have paid more attention to specific data points when forecasting the attacks if they were highlighted in the data.

IV. RESULTS AND DISCUSSION

A. Validation Analysis of Proposed model on Three datasets

In this section, the performance of projected classical is tested with three datasets in terms of different metrics by using 60%-40% and 70%-30% of training and testing data and it is visually shown in Figure 2 to 4.

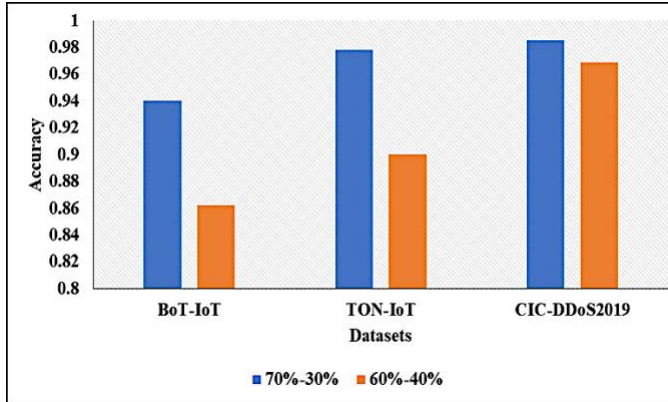


Figure 2: Accuracy Analysis

The analysis of accuracy across two dataset splits, 70%-30% and 60%-40%, reveals notable performance differences. For the BoT-IoT dataset, the 70%-30% split achieves an Accuracy of 0.940, whereas the 60%-40% split achieves 0.862, indicating a drop in accuracy as the training set size decreases. Similarly, for the TON-IoT dataset, the 70%-30% split achieves a higher Accuracy of 0.978 compared to 0.900 for the 60%-40% split, highlighting a significant impact of reduced training data. For the CIC-DDoS2019 dataset, the 70%-30% split achieves the highest accuracy of 0.985, while the 60%-40% split achieves a slightly impressive Accuracy of 0.969. Overall, the results indicate that a larger training set (70%-30% split) consistently leads to better classification accuracy across all datasets.

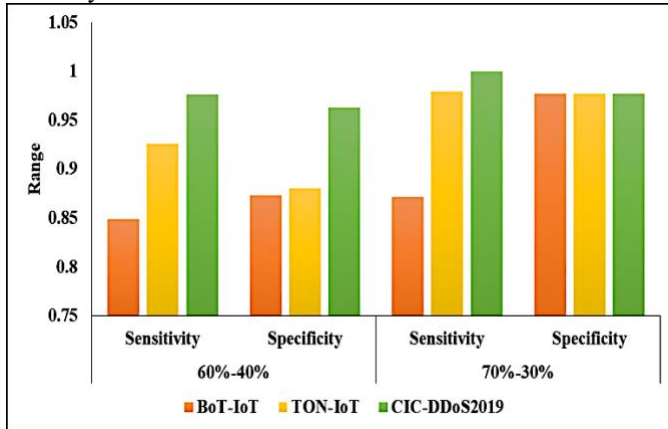


Figure 3: Comparison of three datasets on proposed model

The comparison of sensitivity and specificity for the 60%-40% and 70%-30% dataset splits highlights the performance impact of training data size on classification metrics. For the BoT-IoT dataset, the 70%-30% split achieves higher sensitivity (0.872) and a significantly improved specificity (0.977) compared to the 60%-40% split with sensitivity (0.849) and specificity (0.873). In the TON-IoT dataset, the 70%-30% split outperforms the 60%-40% split, with a remarkable increase in sensitivity (0.979) and specificity (0.977) over the respective values of 0.926 and 0.880 for the smaller training set. The CIC-DDoS2019 dataset shows near-

perfect performance for the 70%-30% split, achieving sensitivity (1.000) and specificity (0.977), compared to sensitivity (0.976) and specificity (0.963) for the 60%-40% split. Overall, the 70%-30% split consistently demonstrates superior sensitivity and specificity across all datasets, reaffirming the benefits of a larger training data size.

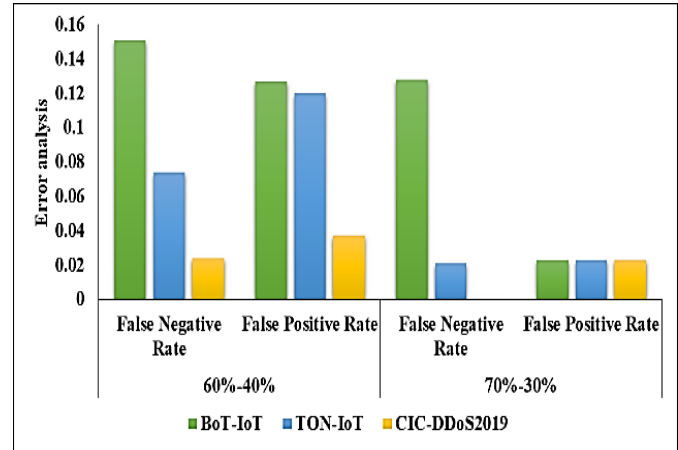


Figure 4: Error analysis

The error analysis comparing FPR for 60%-40% and 70%-30% dataset splits highlights the impact of training data size on error reduction. For the BoT-IoT dataset, the 70%-30% split achieves notable improvements, reducing the FNR from 0.151 to 0.128 and the FPR from 0.127 to 0.023. Similarly, in the TON-IoT dataset, the 70%-30% split drastically lowers the FNR from 0.074 to 0.021 and the FPR from 0.120 to 0.023. The CIC-DDoS2019 dataset shows exceptional results for the 70%-30% split, eliminating FNR (0.000) entirely and reducing the FPR from 0.037 to 0.023. Overall, the 70%-30% split consistently minimizes both error rates across all datasets, underscoring the advantages of larger training datasets for improving model reliability and reducing misclassification errors.

B. Comparative analysis of proposed model

Table 2 presents the comparative study of projected model with existing techniques in terms of training and testing accuracy for combined three datasets.

Table 2: Comparative Study of Projected classical with Existing procedures

Model	Train ACC	Test ACC
AE	0.914 ± 0.0068	0.8687 ± 0.0180
DBN	0.936 ± 0.0019	0.8970 ± 0.0256
GRU	0.920 ± 0.0089	0.8701 ± 0.0342
RNN	0.934 ± 0.0307	0.9063 ± 0.0516
LSTM	0.952 ± 0.0025	0.9102 ± 0.0312
BERT	0.971 ± 0.2763	0.926 ± 0.2280
Ensemble transformer model	0.997 ± 0.0022	0.961 ± 0.0060

Comparative study of the projected model's train and test accuracies against existing techniques, showcasing its superior performance. The Autoencoder (AE) achieves training and testing accuracies of $91.4\% \pm 0.0068$ and $86.87\% \pm 0.0180$, respectively, demonstrating moderate performance. The Deep Belief Network (DBN) and GRU models exhibit slight improvements, with test accuracies of $89.70\% \pm 0.0256$ and $87.01\% \pm 0.0342$, respectively. The RNN and LSTM models perform better, achieving test accuracies of $90.63\% \pm 0.0516$ and $91.02\% \pm 0.0312$. BERT, known for its contextual capabilities, attains test accuracy of $92.6\% \pm 0.2280$, showcasing its robust modeling ability. However, the Ensemble Transformer Model significantly outperforms others with a train accuracy of $99.7\% \pm 0.0022$ and a test accuracy of $96.1\% \pm 0.0060$, underscoring its effectiveness and reliability in handling complex datasets.

V. CONCLUSION

An Anomaly Intrusion Detection System (AIDS) was developed and presented in this research. It is an appropriate option for because to its high efficiency and short detection time. This study offered a unified model to accomplish that. Information gathering, cleansing, feature selection, besides classification are primary processes. A comprehensive preparation process is carried out on the BoT-IoT, TONIoT, and CIC-DDoS2019 datasets to guarantee efficient preprocessing. Then, in order to optimize resource utilisation and feature selection for IoT devices, a BKA strategy is used. Multiple transformers, including BERT, DistilBERT, RoBERTa, and XLNET, are used to classify attacks in the classification step. And to also used SetFit, a framework for few-shot learning improvement in attack classification. Model selection is critical for certain dataset properties, as shown by the results. When all other variables remain constant, RoBERTa's enhanced performance on the standard dataset demonstrates its strong contextual representations. As a result of its efficacy with word augmentation data, BERT is ideal for tasks requiring enhanced inputs. Quick and accurate results are obtained by validating the effectiveness of the suggested model using a tenfold stratified K-fold cross- procedure. There is a lot of room for improvement by investigating and assessing dimensionality reduction methods, even though the suggested method has demonstrated encouraging outcomes. Furthermore, a crucial focus will be on investigating deep learning-based methods, which can extract highly discriminating features. All of these efforts are going towards making the model better at protecting IoT networks and making it more resilient to new types of attacks.

To enhance the presentation of our models, to will finish the hyperparameter tweaking stage in subsequent works. In this step, to tweak the settings of the several machine-learning algorithms to see which ones work best and with what accuracy. To hope to improve the models' learning processes by modifying these hyperparameters, which will increase their predictive capacity and decrease their

overfitting tendencies. A number of metrics, including error, will be used to evaluate the models' efficacy and identify areas for improvement.

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